23CS553 - MINI PROJECT III

“STUDY BUDDY” ACADEMIC LEARNING PLATFORM

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the Study Buddy academic learning platform. This document serves as a reference for developers, testers, project guides, and future contributors throughout the software development life cycle. It clearly explains the expected behaviour of the system, its features, user roles, and performance expectations. The SRS also helps ensure that every module is developed according to the project objectives and provides a common understanding between everyone involved in the design, development, testing, and maintenance of the application.

1.2 Scope

Study Buddy is a web-based learning platform that combines traditional course management with Artificial Intelligence to improve the overall learning experience. The application allows faculty members to create courses, upload study materials, generate quizzes, prepare question papers, and monitor student participation. Students can access learning resources, generate AI-based summaries, practice quizzes, ask questions from uploaded notes, and track their academic progress. The platform is developed using React.js, Node.js, Express.js, MongoDB Atlas, and the Gemini API to provide a secure, responsive, and user-friendly learning environment.

1.3 Definitions, Acronyms and Abbreviations

AI - Artificial Intelligence

API - Application Programming Interface

JWT - JSON Web Token

LMS - Learning Management System

PDF - Portable Document Format

SRS - Software Requirements Specification

MongoDB - NoSQL Document Database

REST API - Representational State Transfer API

Gemini API - Google AI service used for content generation

Cloud Storage – Online storage used for secure document management

1.4 References

- IEEE Std. 830 Software Requirements Specification
- React Documentation
- Node.js Documentation
- Express.js Documentation
- MongoDB Atlas Documentation
- Google Gemini API Documentation

2. Overall Description

2.1 Product Perspective

Study Buddy is a web-based academic learning platform designed to support teaching and learning activities in educational institutions. The system serves as a centralized platform where students, faculty members, and administrators can securely access and manage academic resources based on their respective roles. Faculty members can create and manage courses, upload study materials, prepare quizzes, generate question papers, and monitor student progress. Students can enroll in courses, access learning materials, participate in quizzes, download resources, and track their academic performance through a personalized dashboard. Administrators are responsible for managing users, courses, permissions, and overall platform operations. The application follows a client-server architecture with a React.js frontend, Node.js and Express.js backend, and MongoDB Atlas database, providing a secure, scalable, and modular system that can be extended with additional features in future versions without affecting the existing functionality.

2.2 Product Functions

Study Buddy provides secure user registration and authentication using JWT-based login. Faculty members can create and manage courses, upload PDF study materials, prepare quizzes, assignments, and question papers, and monitor student participation and academic progress. Students can enroll in courses, access and download learning materials, participate in quizzes, view course-related resources, and track their academic performance through a personalized dashboard. The system maintains a history of user activities and generated academic resources, allowing users to access previously created content and download reports

whenever required. Administrators are responsible for managing users, courses, permissions, and monitoring the overall performance and usage of the platform.

2.3 Characteristics of the User

Student

Students can register, log in, enroll in courses, access study materials, can able tosummaries, practice quizzes, ask academic questions, monitor learning progress, and download study resources.

Faculty

Faculty members can manage courses, upload study materials, create quizzes and assignments, generate question papers, monitor student performance, and update course content whenever required.

Administrator

Administrators manage user accounts, assign permissions, monitor system activities, maintain the course database, manage ervice configurations, and ensure the smooth operation of the platform

2.4 Constraints

The platform's document processing features depend on the availability of the Gemini API service. The first version supports only PDF documents for document analysis. Users must have valid login credentials to access system resources. System performance may vary depending on server load and API response time. Regular database backups and security updates are necessary to maintain data integrity, system security, and overall reliability.

2.5 Assumptions and Dependencies

It is assumed that users have basic computer knowledge and access to a modern web browser. Faculty members are expected to upload valid academic content in supported formats. The Gemini API, MongoDB Atlas, and cloud hosting services are assumed to remain available during normal operation. The

system also depends on secure authentication mechanisms, proper server configuration, and periodic software updates to maintain performance, security, and compatibility with future enhancements.3. External Interface Requirements

3.1 User Interface

The Study Buddy platform provides a simple, responsive, and user-friendly interface developed using React.js. The interface is designed to ensure easy navigation for students, faculty members, and administrators. It includes Login and Registration pages, Student Dashboard, Faculty Dashboard, Admin Dashboard, Course Management, Study Material Upload, Smart Notes Summarizer, Automatic Quiz Generator, Question Paper Generator, Study Assistant, Profile Management, Progress Dashboard, and History pages. The interface is optimized for desktops, laptops, and tablets.

3.2 Hardware Interface

- Desktop Computer
- Laptop
- Tablet (optional)
- Smartphone (responsive web access)
- Internet Connection

3.3 Software Interface

- React.js Frontend,HTML,CSS.
- Express.js Backend Framework
- MongoDB Database
- Google Gemini API
- JWT Authentication
- REST APIs
- Modern Web Browser (Chrome, Edge, Firefox)

- REST APIs

3.4 Communication Interface

The platform uses the HTTPS protocol to ensure secure communication between users and the server. Data is exchanged through REST APIs using JSON format. User authentication is managed using JWT tokens, which provide secure access to protected resources. Communication with the Gemini API is performed through secure API requests, while MongoDB Atlas stores application data through encrypted cloud connections. These communication methods help maintain data security, reliability, and efficient system performance.

4. System Features

Study Buddy provides a secure academic learning environment that combines traditional course management with AI-powered educational tools. The platform allows students, faculty members, and administrators to perform their respective tasks efficiently through role-based access. Faculty members can create courses, upload study materials, generate quizzes and question papers, while students can access learning resources, generate summaries, practice quizzes, ask questions based on uploaded notes, and monitor their academic progress. The system maintains a history of AI-generated content and provides downloadable resources whenever required.

5. Other Non-Functional Requirements

5.1 Performance Requirements

The system should respond quickly under normal operating conditions. User authentication should be completed within a few seconds, while AI-based operations such as note summarization, quiz generation, and question paper generation should provide results within an acceptable response time depending on the size of the uploaded document. The platform should support multiple users simultaneously without significant performance degradation. Database queries and file retrieval operations should be optimized to provide a smooth user experience.

5.2 Software System Attributes

The application should provide high reliability through regular database backups and proper error handling mechanisms. User passwords must be encrypted, and all communication between the client and server should be secured using HTTPS and JWT authentication. The modular architecture based on React.js, Node.js, Express.js, and MongoDB Atlas should make the system easy to maintain, test, and expand. AI services should be integrated in a way that allows future upgrades without affecting existing features.

6. Other Requirements

The Study Buddy platform should protect user privacy by securely storing personal information and academic data. Access to study materials and AI-generated content should be limited to authorized users. Regular database backups should be performed to prevent data loss. Future versions of the system may include support for additional file formats, OCR for scanned

documents, multilingual support including Tamil, mobile application support, personalized learning recommendations, and voice-assisted learning features.

Appendix A: Glossary

Term	Description
AI	Artificial Intelligence
LMS	Learning Management System
JWT	JSON Web Token
API	Application Programming Interface
REST API	Representational State Transfer API
PDF	Portable Document Format
MongoDBAtlas	Cloud-hosted NoSQL database
Faculty	User responsible for creating and managing courses
Student	User who accesses learning resources and AI-powered tools